

Data Profiling Report

NYC Yellow Taxi Trip Data (January 2015)

Course: CS-404 Big Data Analytics - Assignment 02

Task 3: Profiling report for the HDFS ingestion and warehousing pipeline

Dataset size profiled: 2,000,000 rows x 19 columns

Team Members: **Ch.Daniyal(453835), Syed Danish(463657), Muhammad Waleed Younas(461215)**

This report summarizes schema structure, missing-value behavior, descriptive statistics, attribute distributions, major data-quality issues, and the proposed cleaning actions to be applied later in the ETL pipeline for Assignment 3.

```
waleed@xyz123: /mnt/c/Users/...
Microsoft Windows [Version 10.0.26280.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pc>wsl
waleed@xyz123:/mnt/c/Users/pc$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [xyz123]
waleed@xyz123:/mnt/c/Users/pc$ stop-yarn.sh
Stopping nodemanagers
Stopping resourcemanager
waleed@xyz123:/mnt/c/Users/pc$ start-dfs.sh
start-yarn.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [xyz123]
Starting resourcemanager
Starting nodemanagers
waleed@xyz123:/mnt/c/Users/pc$ jps
1716 SecondaryNameNode
2134 NameManager
1382 NameNode
1449 DataNode
2520 Jps
1996 ResourceManager
waleed@xyz123:/mnt/c/Users/pc$ cd Desktop
waleed@xyz123:/mnt/c/Users/pc/Desktop$ cd BDA_A2
waleed@xyz123:/mnt/c/Users/pc/Desktop/BDA_A2$ python3 ingest.py
2026-04-19 14:40:20,081 - INFO - === HDFS Ingestion Pipeline Started ===
2026-04-19 14:40:20,082 - INFO - Validating /mnt/c/Users/pc/Desktop/BDA_A2/yellow_tripdata_2015-01.csv
2026-04-19 14:40:20,086 - INFO - File extension validated: .csv
2026-04-19 14:40:20,090 - INFO - File size: 1893.96 MB
2026-04-19 14:40:20,178 - INFO - Detected encoding: ascii (confidence: 100%)
2026-04-19 14:41:37,565 - INFO - Row count: 12748987
2026-04-19 14:41:37,566 - INFO - Validation passed
2026-04-19 14:41:37,567 - INFO - Creating HDFS directory: /warehouse/raw/nyc_taxi/year=2026/month=04
2026-04-19 14:41:40,366 - INFO - HDFS directory ready: /warehouse/raw/nyc_taxi/year=2026/month=04
2026-04-19 14:41:40,367 - INFO - Uploading file: hdfs dfs -put -f /mnt/c/Users/pc/Desktop/BDA_A2/yellow_tripdata_2015-01.csv /warehouse/raw/nyc_taxi/year=2026/month=04/
2026-04-19 14:46:33,029 - INFO - Upload successful to /warehouse/raw/nyc_taxi/year=2026/month=04
2026-04-19 14:46:33,032 - INFO - === Pipeline completed successfully ===
waleed@xyz123:/mnt/c/Users/pc/Desktop/BDA_A2$ python3 profile_data.py
Loading dataset... (This might take a few seconds)

--- SCHEMA DESCRIPTION ---
<class 'pandas.DataFrame'>
RangeIndex: 2000000 entries, 0 to 1999999
Data columns (total 19 columns):
#   Column              Dtype
---  ---
0   VendorID             int64
1   tpep_pickup_datetime str
2   tpep_dropoff_datetime str
3   passenger_count      int64
```

Figure 1. Evidence of HDFS service startup, ingestion execution, and profiling script run.

1. Executive Summary

The January 2015 NYC Yellow Taxi dataset is a strong fit for distributed analytics because it is large, multi-typed, and business rich. The profiled file contains 2,000,000 records and 19 attributes which include vendor identifiers, timestamps, passenger counts, trip distance, geospatial coordinates, payment metadata, and trip charges. The missing-value analysis shows a complete raw file with no null values in any of the 19 columns. The operational value of this system benefits users but does not guarantee absolute quality. The statistical output and the plotted distributions reveal multiple quality concerns: date columns are still stored as strings, several monetary attributes contain negative values, location fields include zero or implausible coordinates, and multiple measures contain very large outliers that are not realistic for normal taxi operations. Most trips look operationally plausible.

Fares are concentrated in the lower range, passenger count is dominated by single-rider trips, and trip distance is strongly right-skewed with most rides appearing short. The extreme maxima for distance and amount variables show that the raw file needs to be cleaned before it can generate trustworthy warehouse facts and dimensional analysis and KPI dashboards. The raw dataset contains complete analytic value and structural integrity but needs specific validation and cleaning work before users can load it into star-schema systems.

2. Dataset Overview

Dataset name: NYC Yellow Taxi Trip Data - January 2015 file (yellow_tripdata_2015-01.csv).

Source used in the project: Kaggle dataset 'NYC Yellow Taxi Trip Data'.

Ingestion evidence supplied by the project run shows a file size of approximately 1.89 GB, detected ASCII encoding, and successful upload to the HDFS path /warehouse/raw/nyc_taxi/year=2026/month=04.

```
waleed@xyz123: /mnt/c/Users/pc$ hdfs dfs -ls -R /warehouse/raw/nyc_taxi/
drwxr-xr-x - waleed supergroup 0 2026-04-18 20:28 /warehouse/raw/nyc_taxi/year=2026
drwxr-xr-x - waleed supergroup 0 2026-04-18 20:56 /warehouse/raw/nyc_taxi/year=2026/month=04
-rw-r--r-- 1 waleed supergroup 1985964692 2026-04-18 20:56 /warehouse/raw/nyc_taxi/year=2026/month=04/yellow_tripdata_2015-01.csv
waleed@xyz123: /mnt/c/Users/pc$
```

Figure 2. HDFS directory structure showing the uploaded raw taxi file.

3. Schema Description

Column	Dtype	Meaning	Representative values
VendorID	int64	Identifier of the taxi vendor/provider	1, 2
tppep_pickup_datetime	str	Trip pickup timestamp	2015-01-15 19:05:39
tppep_dropoff_datetime	str	Trip dropoff timestamp	2015-01-15 19:23:42
passenger_count	int64	Number of passengers in the taxi	1, 2, 3
trip_distance	float64	Distance travelled in miles	0.8, 2.4, 12.7
pickup_longitude	float64	Pickup longitude coordinate	-73.98
pickup_latitude	float64	Pickup latitude coordinate	40.76
RateCodeID	int64	Rate code applied to the trip	1, 2, 5
store_and_fwd_flag	str	Whether record was held in taxi memory before sending	N, Y
dropoff_longitude	float64	Dropoff longitude coordinate	-73.99
dropoff_latitude	float64	Dropoff latitude coordinate	40.74
payment_type	int64	Payment method code	1, 2, 3, 4
fare_amount	float64	Base fare charged for the trip	8.50, 12.00, 52.00
extra	float64	Extra charges such as peak/night surcharges	0.00, 0.50, 1.00
mta_tax	float64	Standard MTA tax amount	0.50

tip_amount	float64	Tip paid by the customer	0.00, 1.50, 2.50
tolls_amount	float64	Tolls charged during the trip	0.00, 5.54
improvement_surcharge	float64	Improvement surcharge amount	0.30
total_amount	float64	Total trip amount paid	10.30, 15.80, 60.00

The profiling output reports 19 columns: 12 float64 fields, 4 int64 fields, and 3 string fields. Representative values below are illustrative examples consistent with the profile and the known taxi schema.

4. Missing Value Analysis

No missing values were reported in the supplied profiling output. Every column has 0 null values, so the raw file is complete from a structural perspective.

Column	Null count	Null %
VendorID	0	0.00%
tpep_pickup_datetime	0	0.00%
tpep_dropoff_datetime	0	0.00%
passenger_count	0	0.00%
trip_distance	0	0.00%
pickup_longitude	0	0.00%
pickup_latitude	0	0.00%
RateCodeID	0	0.00%
store_and_fwd_flag	0	0.00%
dropoff_longitude	0	0.00%
dropoff_latitude	0	0.00%
payment_type	0	0.00%
fare_amount	0	0.00%
extra	0	0.00%
mta_tax	0	0.00%
tip_amount	0	0.00%
tolls_amount	0	0.00%
improvement_surcharge	0	0.00%
total_amount	0	0.00%

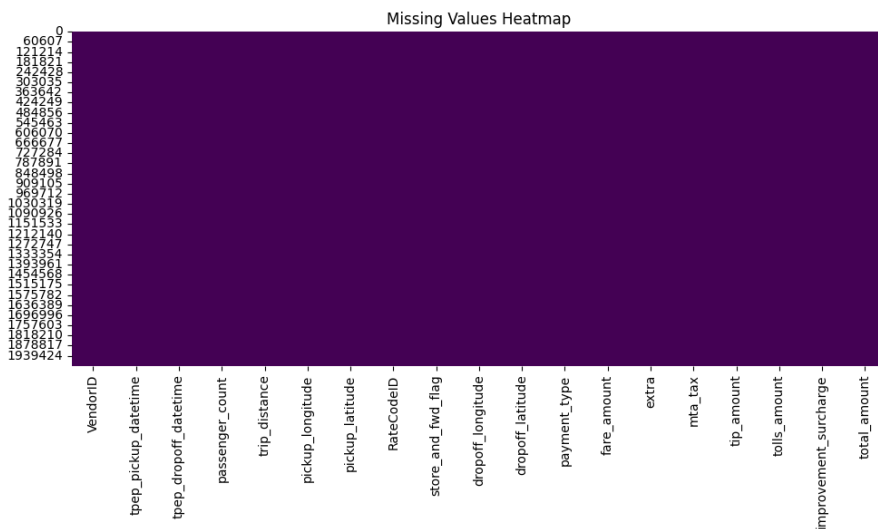


Figure 3. Missing-value heatmap - a solid block confirms that the profile found no nulls across the sampled raw file.

5. Statistical Summary

The descriptive statistics confirm heavy right-skew in key business measures and expose clear anomalies in distance, coordinates, and monetary variables. Values are reproduced from the supplied profile output.

Column	Count	Mean	Std	Min	25%	50%	75%	Max
VendorID	2,000,000	1.522258	0.499504	1.000000	1.000000	2.000000	2.000000	2.000000
passenger_count	2,000,000	1.679528	1.336746	0.000000	1.000000	1.000000	2.000000	9.000000
trip_distance	2,000,000	3.158661	374.365360	0.000000	1.000000	1.660000	3.000000	500501.300000
pickup_longitude	2,000,000	-72.556716	10.143267	-121.925812	-73.991653	-73.981560	-73.966576	0.000000
pickup_latitude	2,000,000	39.970004	5.587742	0.000000	40.735721	40.753300	40.767689	49.389046
RateCodeID	2,000,000	1.036690	0.653185	1.000000	1.000000	1.000000	1.000000	99.000000
dropoff_longitude	2,000,000	-72.603830	9.975178	-89.959541	-73.991219	-73.979767	-73.962517	0.000000
dropoff_latitude	2,000,000	39.996877	5.495273	0.000000	40.731440	40.753685	40.768848	57.470917
payment_type	2,000,000	1.386733	0.498720	1.000000	1.000000	1.000000	2.000000	4.000000
fare_amount	2,000,000	11.889511	10.447217	-450.000000	6.500000	9.000000	13.500000	3005.500000
extra	2,000,000	0.306286	0.368629	-1.000000	0.000000	0.000000	0.500000	100.000000
mta_tax	2,000,000	0.497780	0.035501	-0.500000	0.500000	0.500000	0.500000	0.500000
tip_amount	2,000,000	1.543727	2.417740	-81.000000	0.000000	1.000000	2.660000	750.000000
tolls_amount	2,000,000	0.242630	1.400426	-11.750000	0.000000	0.000000	0.000000	540.330000
improvement_surcharge	2,000,000	0.283839	0.067728	0.000000	0.300000	0.300000	0.300000	0.300000
total_amount	2,000,000	14.779159	12.604892	-450.300000	8.160000	11.150000	16.300000	3006.350000

6. Distribution Analysis

At least five key attributes were visualized. Together, the plots show typical taxi activity patterns mixed with strong skewness and a small number of extreme observations.

Fare amount distribution

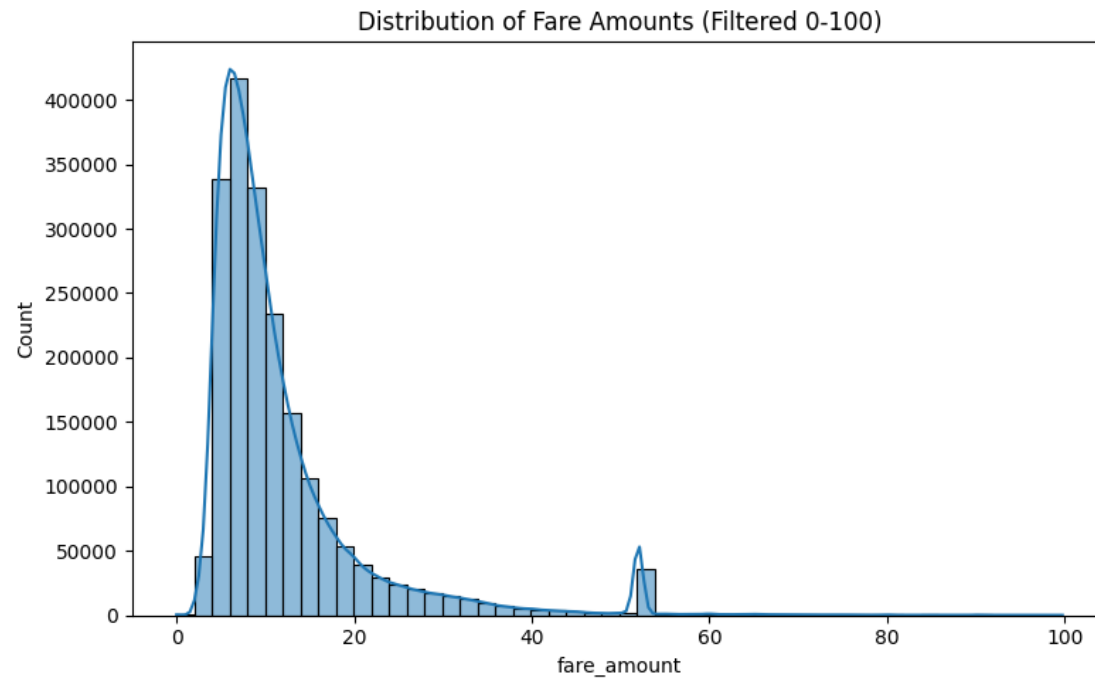


Figure. Fare amount distribution.

Most fares are concentrated roughly between 5 and 15 dollars, with a long right tail. A visible secondary concentration around the low-50-dollar range is consistent with fixed or airport-style charges.

Trip distance distribution

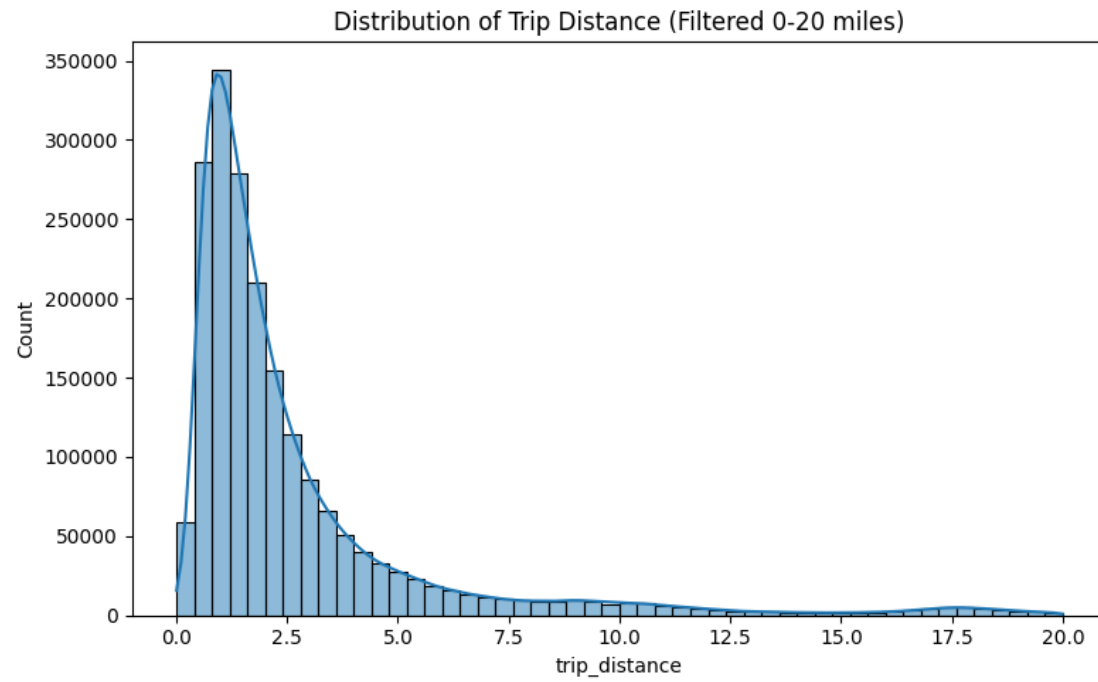


Figure. Trip distance distribution.

Trip distance is strongly right-skewed. Most rides appear short, generally under 3 miles, while a very small set of observations extends far beyond normal taxi behavior.

Tip amount distribution

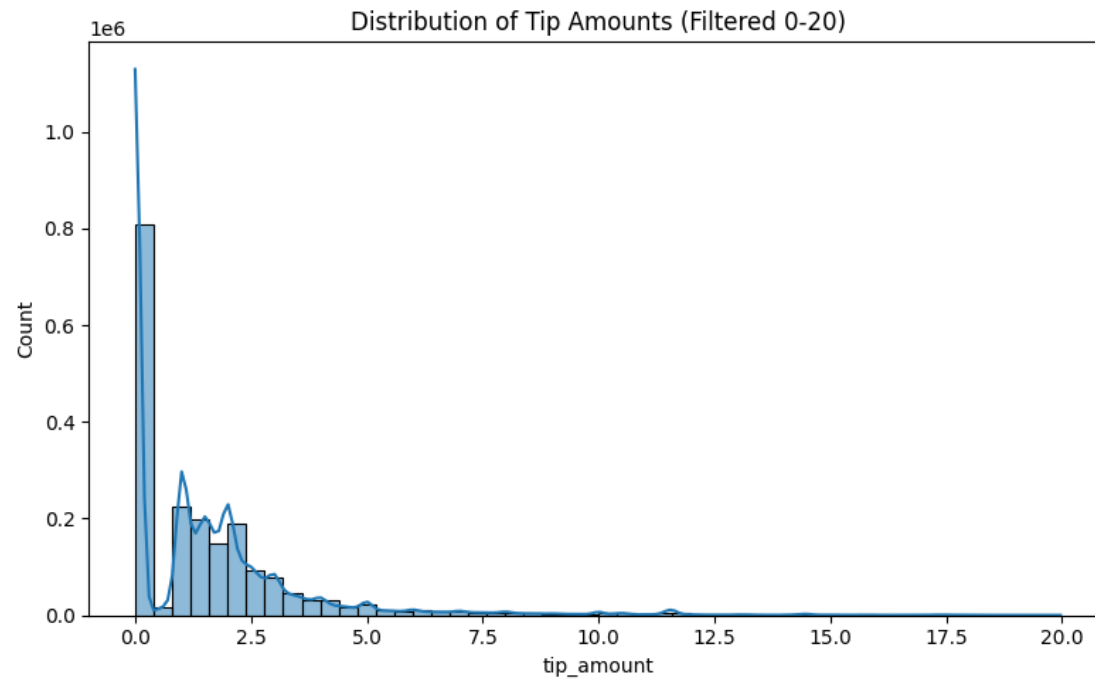


Figure. Tip amount distribution.

A large mass at zero indicates many non-tipped trips or cash trips where tips were not captured in the system. Positive tips cluster at low values and then thin out quickly.

Total amount distribution

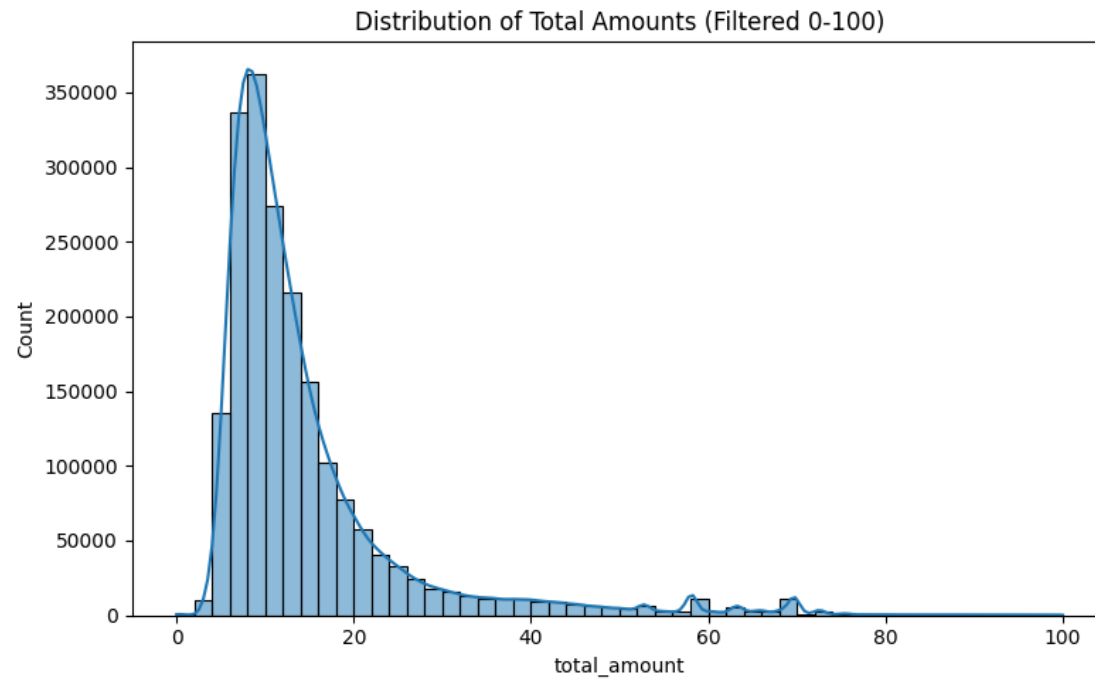


Figure. Total amount distribution.

Total charges are concentrated in the low to mid range, then taper into a long tail. Spikes at certain rounded totals suggest repeated fare rules or fixed-charge patterns.

Passenger count distribution

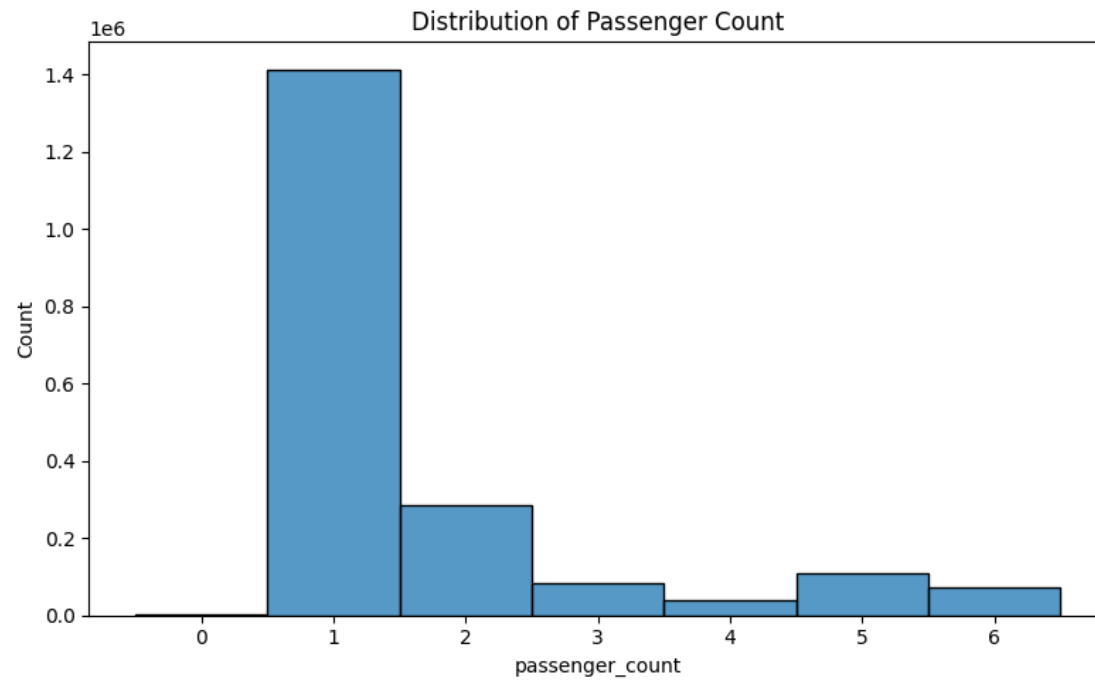


Figure. Passenger count distribution.

Passenger count is dominated by 1-passenger trips, followed by 2-passenger trips. Higher counts are much less common, which aligns with normal city taxi usage.

7. Data Quality Issues Identified

The profile indicates that completeness is good, but validity and consistency still require work. The table below only reports issues that are directly supported by the supplied screenshots and profiling outputs. Where the screenshots do not provide an exact count, the issue is still documented clearly and marked as requiring a code-level count during the ETL stage.

Issue	Evidence from profile	Why it matters	Count / status
No null values	All 19 columns show 0 missing values and 0.00% null rate.	Completeness is strong, so cleaning can focus on validity instead of imputation.	Confirmed
Datetime stored as text	tpep_pickup_datetime and tpep_dropoff_datetime are typed as str in the schema output.	Time analytics, partitioning, and duration calculations require true datetime fields.	2 columns affected
Passenger-count anomalies	passenger_count has min = 0 and max = 9.	A count of 0 is invalid for a completed taxi trip, and very high values require business validation.	Outliers present
Trip-distance anomalies	trip_distance has max = 500,501.3 miles while the median is only 1.66.	This is operationally implausible and will distort aggregates and averages.	Extreme outliers present
Invalid pickup coordinates	pickup_longitude max = 0, min = -121.93; pickup_latitude min = 0, max = 49.39.	Zero and non-NYC coordinates indicate capture or parsing errors.	Invalid values present
Invalid dropoff coordinates	dropoff_longitude max = 0, min = -89.96; dropoff_latitude min = 0, max = 57.47.	Incorrect geospatial fields break zone analysis and mapping.	Invalid values present
Monetary negatives	fare_amount, extra, mta_tax, tip_amount, tolls_amount, and total_amount all have negative minima.	Negative charges are usually entry errors, reversals, or malformed records.	Invalid values present
Monetary extremes	fare_amount max = 3005.5, tip_amount max = 750, total_amount max = 3006.35.	Very large values can heavily bias KPIs and pricing analysis.	Extreme outliers present
Rate-code anomaly	RateCodeID has max = 99 even though normal taxi rate codes are expected to be low integer categories.	Unexpected codes reduce trust in categorical reporting.	Suspicious values present
Duplicate rows	No duplicate count is visible in the supplied screenshots.	Exact duplicates can overstate trips and revenue.	Not evidenced in screenshots; must be counted in code

8. Proposed Cleaning Strategy for Assignment 3

The cleaning strategy below translates each issue into a concrete ETL action. The intent is to preserve as many valid trips as possible while isolating clearly erroneous values.

Issue	Exact cleaning action	Justification	Expected treatment
Datetime as text	Convert both datetime columns with <code>pandas.to_datetime(errors='coerce')</code> . Store parsed values in the warehouse and send unparseable rows to a quarantine table.	Datetime parsing is necessary for trip duration, hourly demand, weekday analysis, and time-based partitions.	Standardize
Passenger count = 0 or implausibly high	Replace <code>passenger_count</code> values outside the valid business range 1-6 with NULL in staging, then impute with the mode passenger count within the same <code>VendorID</code> and hour-of-day bucket. Keep a <code>data_quality_flag</code> .	Passenger count is discrete and heavily concentrated at low values; mode imputation is more appropriate than mean rounding.	Impute + flag
Trip-distance outliers	Keep distances in the normal operating range, but set <code>trip_distance</code> values above a strict upper threshold (for example 100 miles) to NULL and flag the row. Values equal to 0 are retained only when the fare pattern looks plausible; otherwise they are quarantined.	The maximum observed value is impossible for a taxi trip, so it should not enter the fact table as a valid measure.	Cap / quarantine
Invalid coordinates	Set coordinates equal to 0 or outside a defined NYC bounding box to NULL. When both pickup and dropoff coordinates are invalid, keep the financial trip record but exclude the row from geospatial dimension joins.	This preserves revenue facts while preventing invalid map points and zone assignments.	Nullify + retain trip
Negative monetary values	For <code>fare_amount</code> , <code>extra</code> , <code>mta_tax</code> , <code>tip_amount</code> , <code>tolls_amount</code> , and <code>total_amount</code> : convert negative values to NULL in staging. Recalculate <code>total_amount</code> from valid components where possible; otherwise impute component values with the median of a matching trip-distance band and <code>payment_type</code> .	Median-based repair is robust in skewed taxi-charge data and avoids the distortion caused by mean imputation.	Repair
Extreme monetary outliers	Apply business-rule validation plus percentile capping. Values beyond the 99.9th percentile or beyond hard operational limits are moved to an exception table for manual review.	This keeps true premium trips while removing clearly abnormal values from headline analytics.	Winsorize / quarantine
Unexpected <code>RateCodeID</code> values	Map valid low-code categories to a rate dimension and send unknown categories such as 99 to an 'Unknown / Invalid Rate Code' bucket.	This keeps referential integrity without losing the underlying trip record.	Standardize category
Duplicates	Create a full-row hash in staging and remove exact duplicate rows, keeping the first occurrence. Preserve the duplicate count in ETL logs.	Deduplication prevents inflated trip counts and overstated revenue.	Drop exact duplicates
No missing values	No null-value imputation is required in the raw input. The focus should stay on validity checks and derived missingness created during cleaning.	The file is structurally complete, so unnecessary imputation would add noise instead of value.	No action

9. Conclusion

The raw January 2015 taxi file is suitable for warehouse-scale analytics and has strong business value, but it is not analysis-ready in its current form. The most important finding is that this dataset is complete yet noisy: there are no missing values, but there are clear validity issues in timestamps, coordinates, passenger counts, and monetary fields.

The proposed ETL approach therefore prioritizes validation, standardization, targeted imputation, exception handling, and duplicate control. Once these steps are applied in Assignment 3, the data will be much better suited for dimensional modeling, revenue analysis, trip-pattern analysis, and reporting.

Appendix A. Supporting Output Screenshots

```
--- SCHEMA DESCRIPTION ---
<class 'pandas.DataFrame'>
RangeIndex: 2000000 entries, 0 to 1999999
Data columns (total 19 columns):
#   Column              Dtype
---  ---
0   VendorID            int64
1   tpep_pickup_datetime str
2   tpep_dropoff_datetime str
3   passenger_count     int64
4   trip_distance       float64
5   pickup_longitude    float64
6   pickup_latitude     float64
7   RateCodeID          int64
8   store_and_fwd_flag  str
9   dropoff_longitude   float64
10  dropoff_latitude     float64
11  payment_type         int64
12  fare_amount          float64
13  extra                float64
14  mta_tax              float64
15  tip_amount           float64
16  tolls_amount         float64
17  improvement_surcharge float64
18  total_amount         float64
dtypes: float64(12), int64(4), str(3)
memory usage: 289.9 MB
None
```

Schema description output captured from the profiling script.

```

none
--- MISSING VALUE ANALYSIS ---
VendorID          0
tpep_pickup_datetime 0
tpep_dropoff_datetime 0
passenger_count   0
trip_distance      0
pickup_longitude   0
pickup_latitude    0
RateCodeID        0
store_and_fwd_flag 0
dropoff_longitude  0
dropoff_latitude   0
payment_type       0
fare_amount        0
extra              0
mta_tax            0
tip_amount         0
tolls_amount       0
improvement_surcharge 0
total_amount       0
dtype: int64

```

Missing-value count output captured from the profiling script.

```

--- STATISTICAL SUMMARY ---

```

	count	mean	std	min	25%	50%	75%	max
VendorID	2000000.0	1.522258	0.499504	1.000000	1.000000	2.000000	2.000000	2.000000
passenger_count	2000000.0	1.679528	1.336746	0.000000	1.000000	1.000000	2.000000	9.000000
trip_distance	2000000.0	3.158661	374.365360	0.000000	1.000000	1.660000	3.000000	500501.300000
pickup_longitude	2000000.0	-72.556716	10.143267	-121.925812	-73.991653	-73.981560	-73.966576	0.000000
pickup_latitude	2000000.0	39.970004	5.587742	0.000000	40.735721	40.753300	40.767689	49.389046
RateCodeID	2000000.0	1.036690	0.653185	1.000000	1.000000	1.000000	1.000000	99.000000
dropoff_longitude	2000000.0	-72.603830	9.975178	-89.599541	-73.991219	-73.979767	-73.962517	0.000000
dropoff_latitude	2000000.0	39.996877	5.495273	0.000000	40.734440	40.753685	40.768848	57.470917
payment_type	2000000.0	1.386733	0.498720	1.000000	1.000000	1.000000	2.000000	4.000000
fare_amount	2000000.0	11.889511	10.447217	-450.000000	6.500000	9.000000	13.500000	3005.500000
extra	2000000.0	0.306286	0.368629	-1.000000	0.000000	0.000000	0.500000	100.000000
mta_tax	2000000.0	0.497780	0.035501	-0.500000	0.500000	0.500000	0.500000	0.500000
tip_amount	2000000.0	1.543727	2.417740	-81.000000	0.000000	1.000000	2.000000	750.000000
tolls_amount	2000000.0	0.242630	1.400426	-11.750000	0.000000	0.000000	0.000000	540.330000
improvement_surcharge	2000000.0	0.283839	0.067728	0.000000	0.300000	0.300000	0.300000	0.300000
total_amount	2000000.0	14.779159	12.604892	-450.300000	8.160000	11.150000	16.300000	3006.350000

Statistical summary output captured from the profiling script.